Invasive Species, Ecological Context, and Human Impact Assessment

HMCS *Canada* Expedition 2025

HMCS *Canada*/*Queen of Nassau* Deepwater Shipwreck Site, Islamorada, Florida

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Executive Summary

This report provides an integrated invasive species analysis for the HMCS *Canada* shipwreck site (also historically known as the steamship *Queen of Nassau*) located in deep water off Islamorada, Florida, within Florida Keys National Marine Sanctuary (FKNMS). It utilizes HMCS *Canada* Expedition field observations and data to characterize the wreck's biological community and assess the presence of invasive species. Key findings and recommendations emphasize (1) establishing a repeatable baseline of invasive species presence/abundance on the wreck and its adjacent seabed, (2) assessing ecological impacts and pathways that make deep artificial substrates attractive to invaders, and (3) implementing long-term monitoring that links cultural-resource condition, benthic community change, and invasive species dynamics.

1. Background and Management Context

The HMCS *Canada/Queen of Nassau* is a significant cultural resource and artificial reef habitat in the upper mesophotic/deep-reef zone. It lies in approximately 230 feet (≈70 m) of water off Islamorada, Florida, on sandy substrate. The wreck is protected within FKNMS and managed by the National Oceanic and Atmospheric Administration's (NOAA) Office of National Marine Sanctuaries. After its discovery by technical scuba divers in the summer of 2001, ONMS archaeologists conducted photomosaic documentation and broader archaeological study of the site.

In 2025, the HMCS *Canada* Expedition 2025 conducted an interdisciplinary field project aligned with NOAA/ONMS conservation mandates, with objectives that include non-invasive 3D photogrammetric documentation, structural assessment, and marine science components such as ecological assessment and invasive species monitoring. This integrated approach supports mutual benefits: sanctuary management needs (resource protection, condition assessment, and trend detection) and expedition goals (heritage documentation and science contribution).

2. Site Description

Location and setting.

The wreck lies south of Lower Matecumbe Key (Islamorada area) within the sanctuary. The surrounding environment is characterized by deep hard-bottom/mesophotic habitats where light is reduced, temperatures can be more stable than shallow reefs, and currents, generally from the southwest, transport larvae and organic matter to structure.

Physical characteristics.

The site is a ~200-foot steel-hulled shipwreck that rests upright and intact on a sandy bottom. Diagnostic archaeological features include riveted steel construction and a ram bow typical of early 20th-century warship design.

3. Assessment Approach and Data Sources

This analysis uses: (a) NOAA's web resources on the HMCS *Canada*/*Queen of Nassau* wreck and sanctuary invasive species concerns; (b) the expedition's published objectives and location/depth statements; and (c) the user-provided observation that invasive lionfish were found on-site. Where site-specific quantitative values are unavailable, this report identifies data gaps and provides NOAA-consistent monitoring designs to produce defensible metrics in future sampling.

4. Invasive Species Risk Profile for the Wreck

NOAA identifies non-indigenous species as a major threat to ecosystem integrity due to their ability to alter community composition, reduce native diversity, interfere with ecosystem function, and once established, become difficult to eradicate. In the Florida Keys, documented non-indigenous taxa of management interest include Indo-Pacific lionfish, two species of the non-indigenous orange cup coral (*Tubastraea coccinea; T. tagusensis*), the latter noted as having expanded offshore of Key Largo since 1999 but remaining largely confined to artificial substrates. NOAA also notes seasonal abundance of the Red-tipped Sea Goddess nudibranch (*Glossodoris sedna*), with uncertain ecological effects. There are also multiple species of non-native bivalves that have been identified on shipwrecks, at low abundance; ecological effects of these are also unknown. More recently, three Indo-Pacific soft coral species commonly referred to as "pulsing corals" (family Xenidae; Xenia, Unomia and Latissimia spp.) have been spreading through the wider Caribbean, Hawaii. They were first noted off Venezuela but are now spreading through reef habitats and on artificial substrates in Puerto Rico, Cuba and elsewhere; they have not yet appeared in Florida. They are of major ecological concern because of their rapid growth rates, efficient asexual dispersal methods, and lack of predators, and they have been found to displace local benthic organisms, reduce local biodiversity and expand into habitat types outside of typical coral reef habitats.

4.1 Invasive Lionfish on Deep Wrecks

Presence at HMCS *Canada*.

The HMCS *Canada* Expedition found that many lionfish were present on the HMCS *Canada* wreck. The number of lionfish was much higher than found on a nearby shipwreck (a site known as the Cannabis Cruiser) that rests at a depth of 100 feet and is accessible to recreational divers for lionfish culling. ONMS visits in 2001, 2002, 2003, and 2005 did not encounter lionfish at HMCS *Canada*/*Queen of Nassau*.

Why lionfish occur on deep wrecks.

Lionfish are habitat generalists that occupy both natural reefs and artificial structures (including shipwrecks) across a broad depth range. Artificial reefs often provide: (1) complex refuge space that increases hunting success and reduces predation risk; (2) concentrated prey communities that aggregate around structure; and (3) stable hard substrate in otherwise sandy environments that create "islands" of habitat readily colonized by larvae.

Ecological impacts and damage potential.

NOAA reports that lionfish have very few predators in invaded ranges and can substantially depress native reef fish recruitment; NOAA materials cite a 79% reduction in recruitment of native reef fish associated with a single lionfish on a reef in experimental settings. By reducing small-bodied reef fish abundance and altering trophic pathways, lionfish can diminish ecosystem resilience and indirectly affect fisheries-relevant species through prey depletion and competition (NOAA Fisheries).

Operational constraints for deep sites.

At ~70 m depth, dive time is limited and removal/collection approaches must prioritize diver safety. This makes deep wrecks potential refuges from control programs, emphasizing the value of periodic targeted removals during planned scientific/archaeological dives and consideration of emerging deep-removal tools (e.g., traps) where permitted and appropriate.

4.2 Non-indigenous Orange Cup Coral (*Tubastraea* spp.) and Other Fouling Invaders

Orange cup coral (*Tubastraea coccinea*) is a non-indigenous, ahermatypic (non-reef-building) coral that can colonize hard substrates such as shipwrecks, pilings, caves, and rigs. NOAA Florida Keys materials note that orange cup coral has expanded in offshore areas since the late 1990s and has thus far been largely confined to artificial substrates—making wrecks a priority habitat for early detection and mapping. Because this coral is azooxanthellate (does not rely on symbiotic algae), low-light deep wreck conditions are not necessarily limiting, potentially enabling persistence and spread on shaded surfaces and interiors. It is unclear at this time if these corals are outcompeting native species.

There were no confirmed *Tubastraea* observations at HMCS *Canada*/*Queen of Nassau* in 2025; however, its colonization was anticipated due to documented regional occurrence on artificial substrates. Recommended future action would be to include *Tubastraea* in photoquadrat and photogrammetry annotation protocols and, if detected, to map colony distribution (hull vs interior, shaded vs exposed surfaces) and assess potential competitive interactions with native sponges, octocorals, and other sessile taxa.

5. Why Deep Wrecks Attract and Sustain Invaders

Deep wrecks function as persistent, high-relief hard substrate in a region where surrounding seabed may be predominantly sand or low-relief hardbottom. This enhances settlement opportunities for sessile organisms and creates a concentrated food web that supports predators such as lionfish. NOAA identifies multiple introduction pathways relevant to the marine environment, including shipping/ballast water, hull fouling/transfer, and accidental or intentional releases; once established, deep sites may experience lower human removal pressure and can serve as source populations for recolonization of other habitats.

6. Ecological Context: What Is Typically Found on Deep Shipwreck Sites

Upper-mesophotic/deep reef shipwrecks commonly host mixed communities of sponges, octocorals/soft corals, stony corals tolerant of lower light, bryozoans, hydroids, and encrusting algae, along with mobile invertebrates (e.g., crustaceans) and aggregations of reef-associated fishes. Structural complexity supports sheltering and foraging, and wrecks may act as stepping-stones for species dispersal across depth gradients. For heritage management, these biological communities can both stabilize and accelerate deterioration: encrustation may shield surfaces from direct exposure, while biofouling and localized corrosion processes can contribute to material loss over time.

7. Monitoring Importance and Integrated Metrics

Monitoring HMCS *Canada* is important for both resource protection and ecosystem science. As a deep cultural resource, the wreck provides a fixed reference point for detecting change in benthic communities and invasive species distributions through time. Integrated monitoring supports NOAA's mandate to manage natural and cultural resources together, enabling managers to link biological change to physical impacts (e.g., fishing gear interactions) and environmental stressors.

8. Recommended Monitoring and Analysis Framework

8.1 Baseline products (Year 1).

- 3D photogrammetric model annotated with habitat zones (deck, hull, interior openings, debris field) and sessile cover classes.
- Invasive species inventory with photographic vouchers and, where permissible, limited specimen confirmation.
- Quantitative abundance estimates for lionfish (counts per standardized survey time/area) and sessile invaders (percent cover or colony counts using roving or ROV surveys).

8.2 Repeat sampling design (Annual or Biennial).

- Fixed photo stations covering areas with species present and representative control areas to document spread (paired oblique + nadir images).
- Standardized lionfish surveys at consistent times of day; record size class (estimated total length) to infer population structure.
- If removals occur, track effort (diver-minutes), catch per unit effort (CPUE), and recolonization rates.

8.3 Data integration and reporting.

- Integrate biological metrics with historical-resource condition indicators (structural stability, corrosion hotspots, fishing gear interactions).

- Maintain a shared metadata standard (site datum, coordinate reference, camera calibration, visibility, currents) so NOAA and partners can compare across years.
- Publish a concise annual condition-and-invasive summary for sanctuary managers and expedition partners.

8.4 Triggers for management action.

- Lionfish density exceeding locally defined thresholds or repeated observations of juveniles indicating on-site recruitment.
- Detection of *Tubastraea* spp. or other high-concern sessile invaders appearance on the wreck.
- Evidence of ecological shift (loss of native small-bodied fishes, increased bare substrate) potentially linked to invasive predation or competition.

9. Data Gaps and Priority Needs

Priority gaps for a comprehensive NOAA-grade assessment include:

1. a full species list with abundance/cover estimates for the wreck and surrounding seabed;
2. repeated measurements that separate seasonal variability from directional change;
3. documentation of fishing gear interactions and any evidence of anchoring impacts;
4. environmental context (temperature, dissolved oxygen, currents) to interpret biological patterns; and
5. a centralized archive for imagery, models, and annotations.

HMCS *Canada* Characterization
Dive Date: December 1 - 4, 2025
Dive Team: Joseph Frey, Guy Shockley, Ewan Anderson, Jason Cook, Rob DePoy, Dr. Roger Lacasse, Kelvin Davidson. Characterization author(s) through video, and imagery: Lauri MacLaughlin, Brenda Altmeier, Andy Bruckner, Ewan Anderson

Species Identified:

Organism	Genus / Species
Stony Corals	
Smooth Starlet Coral	*Siderastrea siderea*
Sponges	
Branching Vase Sponge	*Callyspongia vaginalis*
Algae	
Red Algae	*Peyssonnelia*
Tunicates	
None Identified	
Invertebrates	
Spiny Oyster	*Spondylus americanus*
Fish	
Lionfish	*Pterois miles* 22 +
Atlantic Goliath Grouper	*Epinephelus itajara* *3*
Reef Butterflyfish	*Chaetodon sedentarius*
Spotfin Hogfish or Cuban Hogfish	*Bodianus pulchellus*
Blue Runner	*Caranx crysos*
Almaco Jack	*Seriola rivoliana*
Striped Grunt	*Haemulon striatum*
Creole Fish	*Paranthias furcifer*
Tomtate	*Haemulon aurolineatum*
White Grunt (Juvenile and Adult)	*Haemulon plumierii*
Sunshine Fish	*Chromis insolata*
Boga	*Haemulon vittatum*
Bar Jack	*Caranx ruber*
Great Barracuda	*Sphyraena barracuda*

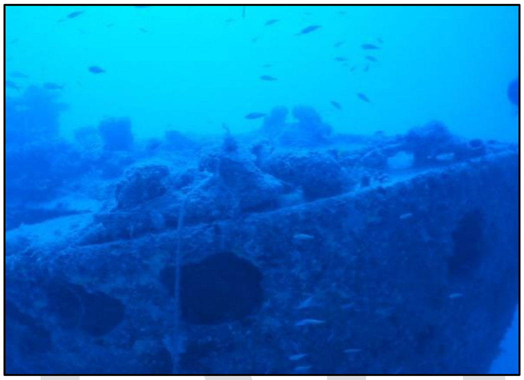
The HMCS *Canada* Expedition team encountered several Atlantic goliath groupers on the shipwreck. Considered for inclusion under the U.S. Endangered Species Act, but not yet listed, the fish is considered a species of concern due to exploitation. Harvest of goliath grouper was prohibited in state and U.S. federal waters beginning in 1993. The National Marine Fisheries Service continues to list the goliath grouper as overfished. Goliath grouper are slow growing and have predictable spawning aggregations in August and September. The largest individuals represent the greatest migrations to spawning sites. After spawning the species moves back to their respective home sites in October and November (Ellis et al. 2023).

Large goliath grouper encountered by the dive team. Photo: Ewan Anderson, HMCS *Canada* Expedition

Indo-Pacific lionfish observed during the 2025 HMCS *Canada* Expedition. Photo Ewan Anderson

4.1 Invasive Lionfish on Deep Wrecks
Presence at HMCS *Canada*.

Video footage taken during the field project (20251203.mp4) by Ewan Anderson.
Video review beginning at HMCS *Canada* port bow starting time code 2:07 through to video time code 2:50, video ending approximately port amidship, depths 100+ feet, there was an estimated count of 22 lionfish observed while evaluating footage.


Frame 2:07


Frame 2:18


Frame 2:28


Frame 2:44

Human-Use Impacts Observed at the HMCS *Canada* Deep-Water Shipwreck Site

The HMCS *Canada*/*Queen of Nassau* is a focal point for human activity in the offshore environment. Expedition imagery revealed anthropogenic impacts associated primarily with fishing and boating activities. These impacts are consistent with patterns documented on other deep artificial reef structures in the Florida Keys region and represent both direct physical stressors to the wreck and secondary ecological pressures on associated benthic communities.

1. Derelict Fishing Gear ("Ghost Gear")

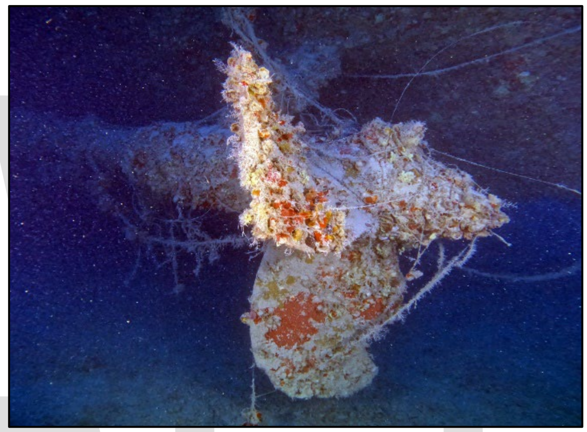
Imagery documented multiple instances of:

- Synthetic netting fragments snagged on protruding wreck structures
- Monofilament fishing line draped across railings, deck features, and debris field elements
- Hooks, sinkers, and terminal tackle hanging from nearly all structures

- Synthetic polyline, commonly used in lobster trap systems entangled on wreck structures.

Biological Implications:

- Continual entanglement risk to non-target species including sharks, sea turtles, and mobile invertebrates
- Ongoing "ghost fishing" mortality where nets or lines continue capturing organisms long after loss
- Physical abrasion and breakage of sponges, octocorals, and encrusting communities
- Accelerated corrosion where synthetic lines trap sediments and organic material against metal surfaces



Monofilament line and fishing lure caught on shipwreck. Photo Ewan Anderson

Net draped across the ship structure. Photo: Ewan Anderson

Left – trap line entangled in ship structure. Right – trap line entangled in artifact.
Photos: Ewan Anderson

2. Anchoring

Imagery documented multiple instances of:

- Anchor entanglement with wreck superstructure and artifacts
- Anchor rode entanglement with wreck superstructure and artifacts

Biological Implications:

- Crushing of benthic organisms
- Destabilization of structures used for sessile organism attachment
- Abrading biological growth during current movement

Modern anchor and chain entangled in ship structure. Photo: Ewan Anderson

Broader Ecological Consequences

The combined effects of fishing and boating debris create a feedback loop of degradation:

Impact Mechanism	Ecological Effect	Cultural Resource Effect
Entanglement	Mortality of reef organisms	Visual obstruction & structural stress
Abrasion	Loss of sponges/corals	Accelerated corrosion
Sediment trapping	Smothering of benthos	Micro-environmental decay

Deep-water communities grow more slowly than shallow reef systems, meaning damage persists longer and recovery is significantly delayed.

Relevance to Invasive Species Dynamics

Human debris can inadvertently facilitate invasive species establishment by:

- Creating microhabitats favorable to non-native sessile organisms (e.g., cup corals)
- Providing refuge structures that concentrate prey for lionfish
- Reducing native species resilience through physical stress

Thus, fishing and boating impacts indirectly amplify invasive species pressures already documented at the site.

Management and Monitoring Significance

The HMCS *Canada* Expedition's integration of biological survey with heritage documentation is particularly valuable because:

- It captures both ecological change and human-use stressors in the same dataset
- Provides visual evidence necessary for targeted mitigation (gear removal, outreach, mooring evaluation)
- Establishes baseline conditions against which future impact trends can be measured

Recommended Impact Metrics for Future Monitoring

To complement invasive species tracking, future surveys should quantify:

Fishing & Debris Indicators

- Number of net fragments per survey zone
- Linear meters of fishing line present
- Tackle density (hooks/sinkers per area)
- Trap line contact points

Removal Tracking

- Gear type removed
- Removal effort (time/divers)
- Re-accumulation rate

Summary Statement for NOAA and Partners

Visual evidence from the 2025 interdisciplinary survey of the HMCS *Canada* shipwreck demonstrates that fishing and boating activities represent significant ongoing stressors to both the historical resource and its associated deep-reef ecosystem. Lost netting, fishing lines, terminal tackle, anchor interactions, and trap line were observed entangled on wreck structures and surrounding seabed. These materials contribute to chronic entanglement risk, physical abrasion of benthic communities, sediment trapping, and accelerated corrosion of metal features. Beyond direct damage, such impacts may indirectly facilitate invasive species persistence by reducing native community resilience and creating favorable microhabitats. Integrated monitoring of human-use impacts alongside invasive species dynamics is therefore essential for effective long-term management of this culturally and ecologically significant site.

References Cited

NOAA Fisheries
2025 Impacts of Invasive Lionfish.
https://www.fisheries.noaa.gov/southeast/ecosystems/impacts-invasive-lionfish, website accessed 11 February 2026

Ellis, Robert D. Christopher C. Koenig, James V. Locascio, Christopher R. Malinowski, and Felicia C. Coleman
2023 Spawning Migrations of the Atlantic Goliath Grouper along the Florida Atlantic Coast. *Fishes*, 8(8), 398
https://repository.library.noaa.gov/gsearch?parentId=&terms=Spawning+Migrations+of+the+Atlantic+Goliath+Grouper+along+...
accessed 12 February 2026
DOI : https://doi.org/10.3390/fishes8080398